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IT and IS Department

Course Code	ITCD 103	Section	29104
Course Details	Software Development for Enterprise Systems	Date	02/08/26
Names	Allan Mark Gardoce	Hands-On Laboratory	

Chapter 2: Understanding Enterprise Architecture
Hands-on Laboratory Exercise 2

Objective: This laboratory exercise aims to enhance your understanding of Customer Relationship Management (CRM) systems and their components by designing a detailed diagram using Draw.io. Through this exercise, you will:

1. Identify and visualize the key modules and functionalities of a CRM system.
2. Demonstrate how different components interact and exchange data.
3. Represent user roles and access control within the system.
4. Gain practical experience using Draw.io for diagramming complex systems.

Lab Environment: This exercise assumes access to a computer with internet connectivity and Draw.io software or Desktop Software. Basic knowledge of CRM systems and Draw.io is recommended.

Tasks:

1. **Choose a CRM System:** Select a specific CRM platform you are familiar with or a generic CRM you'd like to model.
2. **Research Key Features:** Explore the chosen CRM's website or documentation to understand its core modules, functionalities, and integrations.
3. **Design the Diagram:**
    - Start with the essential components like Contacts, Sales, Marketing, Service, Reporting, and analytics.
    - Expand each module with sub-components and specific features.
    - Use arrows and lines to show data flow and interactions between components.
    - Represent external systems interacting with the CRM.
    - Differentiate user roles and access levels within modules.
    - Include additional details like business processes, KPIs, and security considerations (optional).
4. **Annotate and Style:**
    - Add clear labels and brief explanations for each component and functionality.
    - Use consistent colors, fonts, and shapes for better organization and visual appeal.

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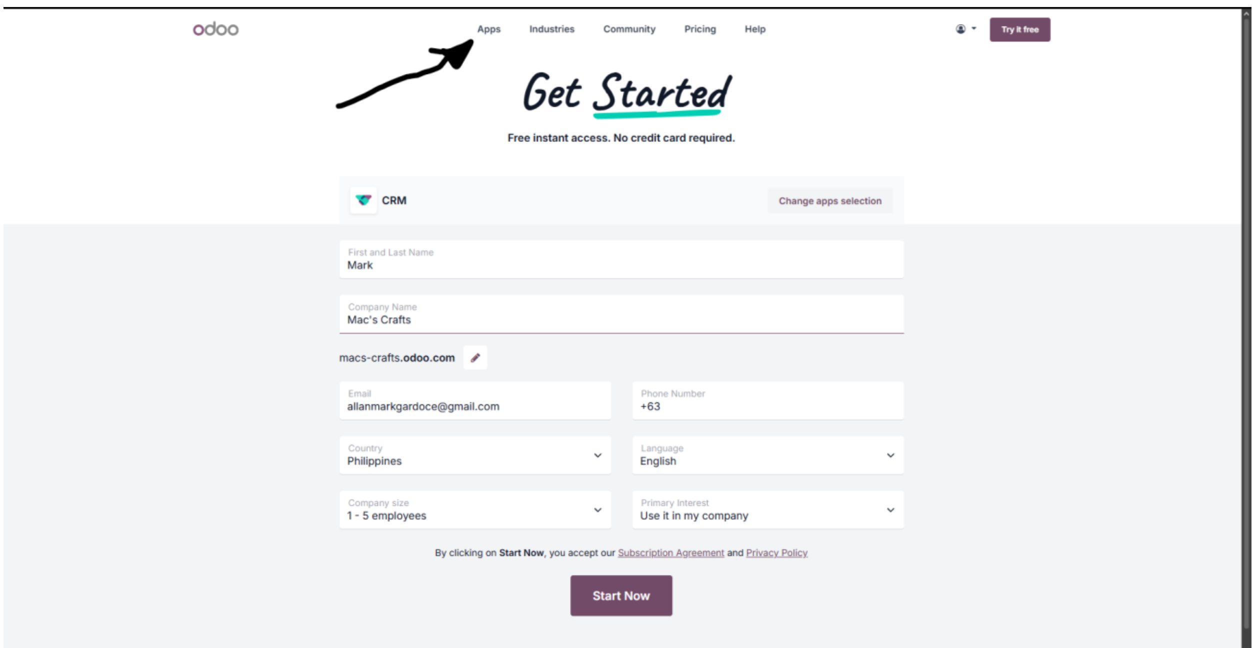
Conclusion:

- This laboratory exercise successfully enhanced my understanding of how Enterprise System Architecture integrates diverse business modules into a unified, secure environment. By modeling the Odoo CRM, I visualized how modular architecture facilitates seamless data exchange between Marketing and Sales while maintaining a "Single Source of Truth" within a centralized database. Furthermore, representing user roles and access levels underscored the critical importance of security and permission hierarchies in modern organizational software. Ultimately, this hands-on engagement with Draw.io and the Odoo ecosystem has provided practical experience in analyzing and diagramming the complex interdependencies that drive enterprise-level operations.

Print Screen the Visual Representaion:



- This visual captures the initial stage of exploring the **Odoo ecosystem**, where the **"Apps"** menu serves as the primary gateway to the system's modular architecture. Navigating to this section is a fundamental step in identifying the various functional components available within the enterprise environment, allowing for a high-level overview of how different business applications are categorized and managed.

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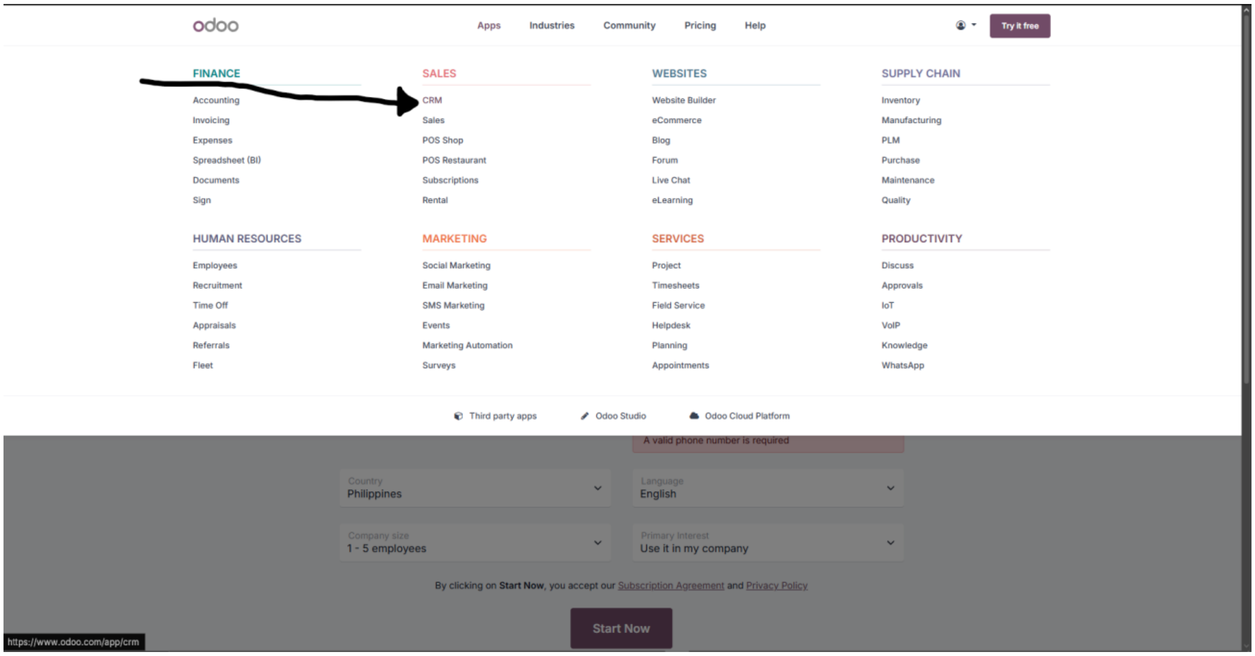
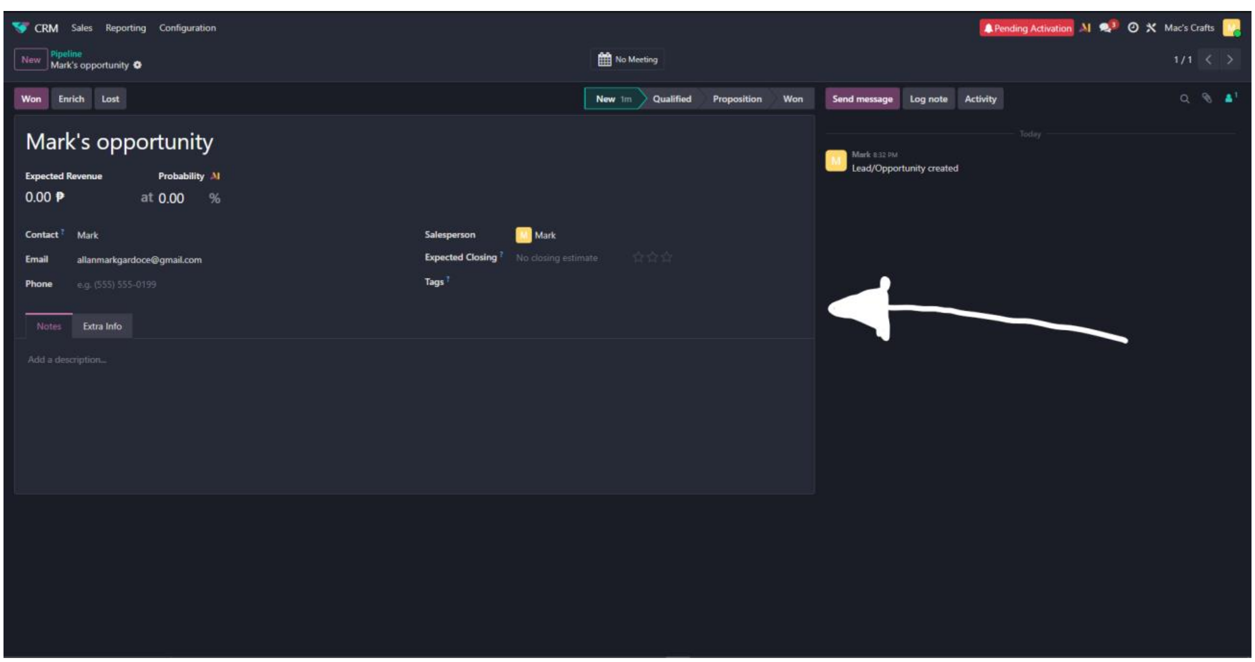
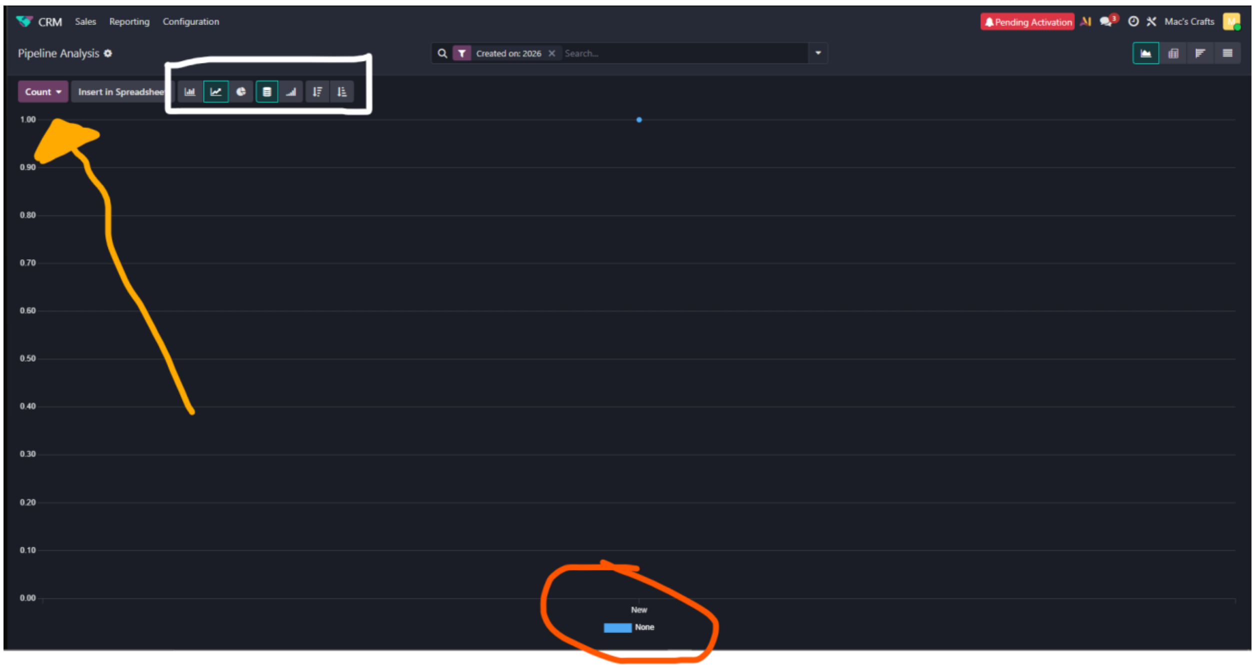


- In this screenshot, I have isolated the **CRM module** within the broader Sales category, identifying it as the foundational engine for managing customer relationships and lead lifecycles. This selection is critical for researching the system's core functionalities and understanding how this specific component interacts with adjacent modules, such as **Marketing** or **Accounting**, to facilitate seamless data exchange across the enterprise



- The Odoo CRM architecture integrates diverse business functionalities through a stage-based pipeline, highlighted by the white rectangle in the screenshot, which serves as the foundational logic for guiding sales opportunities from initial entry to successful completion. This visual demonstration of the business process flow confirms that the enterprise environment enforces standardized workflows to maintain operational consistency and provide real-time data for accurate financial forecasting. Furthermore, the yellow arrow illustrates the relational nature of the system's database, where associating a specific contact automatically retrieves linked information like email addresses from a centralized master record, effectively demonstrating the "Single Source of Truth" principle essential for data integrity. Finally, the pink circle identifies the system's integrated audit trail through the Log Note feature, which ensures organizational transparency and accountability by maintaining a permanent, chronological record of all user interactions within a specific business object. Together, these features highlight an architecture designed for seamless data exchange, automated process management, and secure operational tracking



- In the final task, I utilized the **Odoo Reporting** engine to evaluate how the system architecture processes raw data into meaningful analytics. By accessing the **Pipeline Analysis** dashboard, I observed that the system automatically aggregates disparate records from the CRM and Sales modules to present real-time KPIs, such as the total count of active leads. The availability of dynamic visualization tools—including bar, line, and pie charts—demonstrates the system's role as a powerful decision-support layer that provides stakeholders with a high-level overview of organizational health. This experience confirmed that the true value of an Enterprise System lies in its capacity to transform static database entries into the visual business intelligence required for strategic, data-driven planning.

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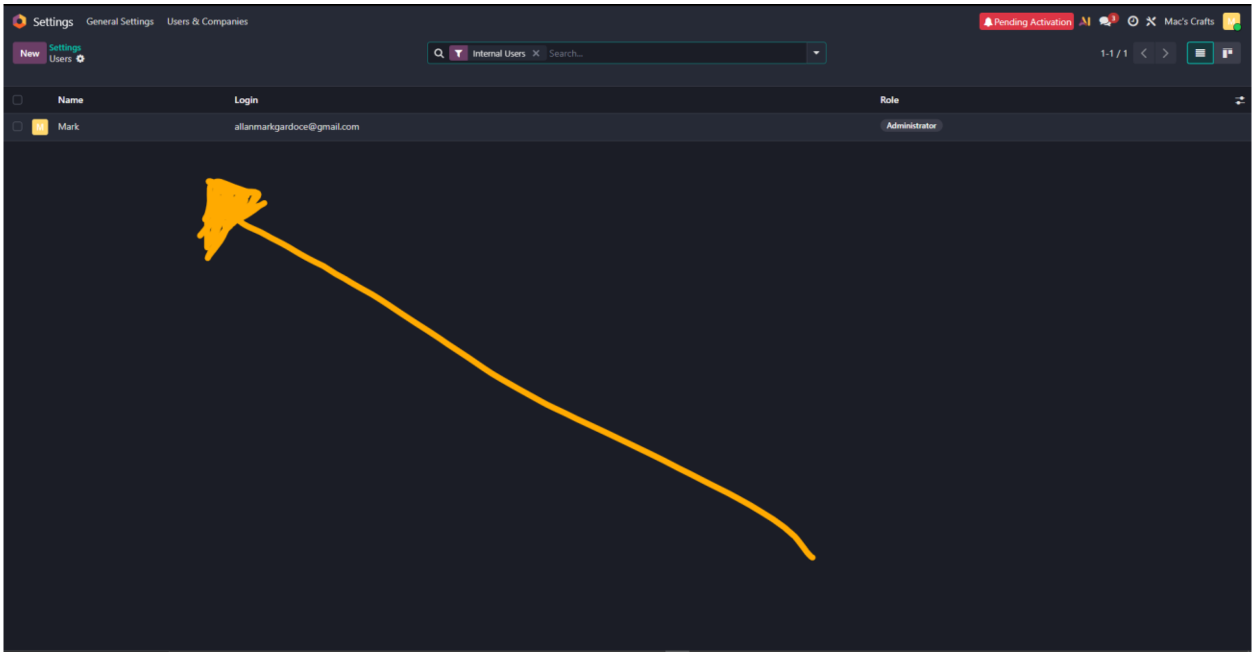
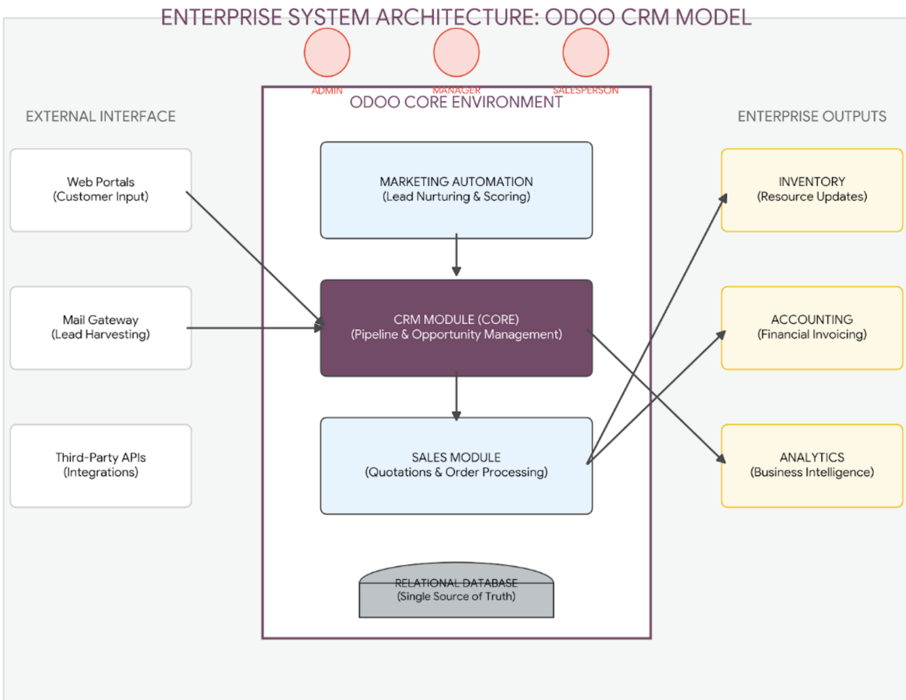


- This final visual captures the **Security and Access Control** framework of the Odoo architecture, specifically highlighting the management of user roles and functional permissions. By examining the detailed access rights for the administrative profile, I observed how the system partitions authority across diverse modules such as Sales, Marketing, and Master Data management. This hierarchical security model is essential for maintaining data integrity and confidentiality within an enterprise environment, as it ensures that sensitive organizational information is only accessible to authorized personnel based on their specific business responsibilities.

DIAGRAM:



- **External Interface:** This layer represents the "ingestion points" where data enters the system from the outside world. Web portals and mail gateways act as automated sensors that capture lead information and funnel it directly into the CRM.
- **Odoo Core Environment:** The central purple and blue boxes represent the primary modular architecture. The **CRM Module** acts as the command center, receiving scored leads from **Marketing** and converting them into actionable opportunities for the **Sales Module**.
- **Relational Database:** Positioned at the foundation, this represents the **"Single Source of Truth."** It ensures that every module—from Accounting to Inventory—accesses the same synchronized contact record, eliminating data redundancy.
- **Enterprise Outputs:** These are the "downstream" results of a successful sales workflow. A closed opportunity in the CRM automatically triggers stock updates in **Inventory** and invoice generation in **Accounting**, while providing the raw data for **Business Intelligence (Analytics)**.
- **User Security Layer:** The icons at the top illustrate the **Access Control** hierarchy. Each role (Admin, Manager, Salesperson) has distinct "keys" to the system, ensuring that sensitive financial or master data is only accessible to authorized personnel.

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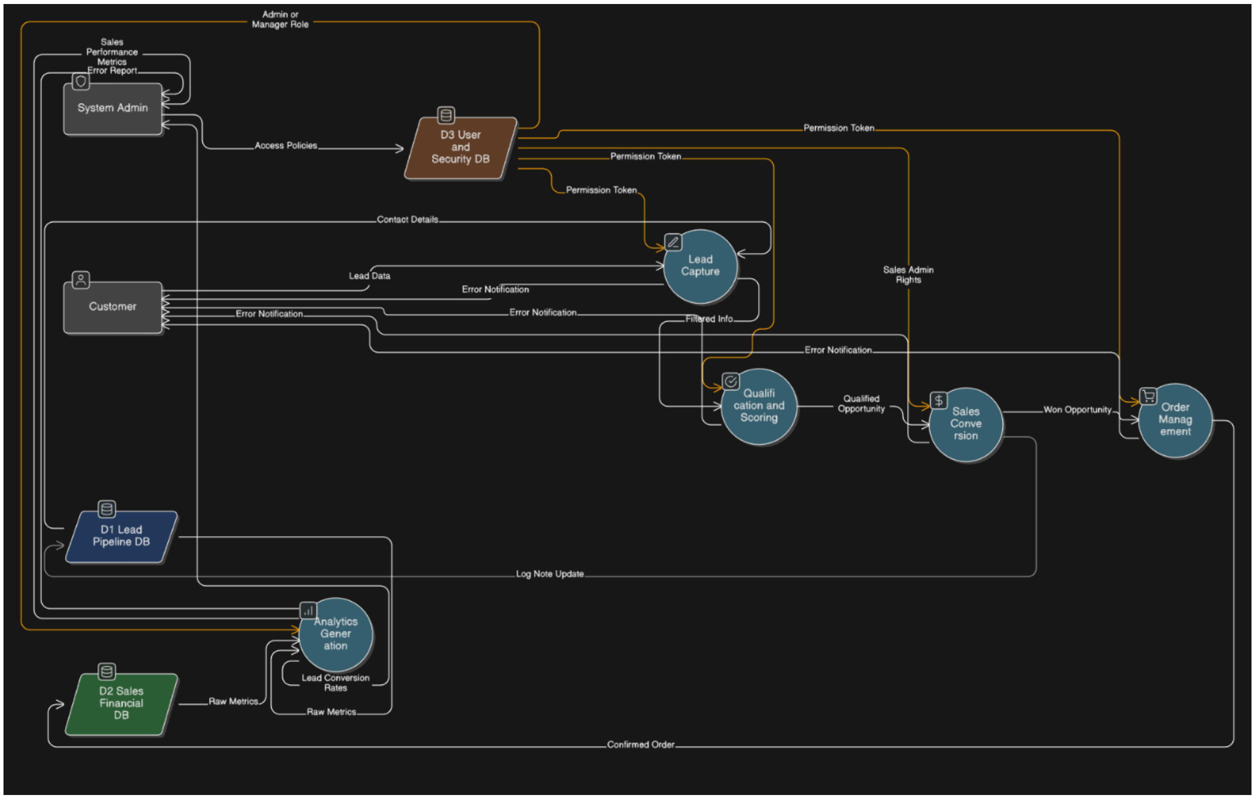
DFD:



This Data Flow Diagram (DFD) illustrates the logical processing of data within the Odoo CRM ecosystem. It highlights how raw external inputs are transformed into actionable business intelligence through a series of governed processes and centralized data stores.

- Process 1.0 (Lead Capture & Filtering): Acts as the primary gateway, ingesting Inbound Lead Data from external entities like Customers. It ensures data integrity by filtering inputs before they enter the core system.
- Process 2.0 (Qualification & Scoring): This process interfaces with D1 (Lead/Pipeline DB) to rank opportunities. It establishes the relational link between individual leads and organizational sales targets.
- Process 3.0 & 4.0 (Conversion & Order Management): Represents the transition from a prospect to a confirmed business transaction. Once a lead is "Won," the data flow updates D2 (Sales/Financial DB), which serves as the organization's Single Source of Truth for revenue.
- Process 5.0 (Analytics Generation): This process "reads" the accumulated data from D1 and D2 to output visual KPIs. It provides the necessary business intelligence for strategic decision-making by management.
- D3 (User & Security DB): This data store governs all system interactions. By feeding Access Policies into the processes, it ensures that only authorized roles, such as the Administrator, can execute specific functions like sales conversion or report generation.